

K473 — Cal #347 Catches K471 Over-Tiers (Second Cal Audit on Keeper in Two Days); Walk-Back Clean; K471 Headlines Retiered Honestly

Date: 2026-06-22 (Monday, ~12:30 EDT date-verified) · Auditor: Keeper (self-audit absorbing Cal #347) · Inputs: Cal Referee Log #347 catching four over-tiers in K471 roll-up

Verdict — Cal #347 catches me cleanly; K471 elevated four things to SOLID/Paper-grade ahead of evidence; walk-back clean. SECOND Cal-on-Keeper audit event in two days. Pattern flag.

Count holds at 4 of 26. The substantive content this turn: Cal has caught Keeper TWICE in two days for the same pattern — taking the team’s substantive bench work and over-tiering it in roll-ups.

Cal #347’s four specific catches

Catch 1 — The Lyra/Elie 0^{-+} contradiction K471 cited as “support”

Cal flagged the internal contradiction: - Lyra: $0^{-+} = \text{Tr}(F \cdot \tilde{F})$ — quadratic, SAME color block, differs by spacetime Hodge dual - Elie 4308: $0^{-+} = d^{\{abc\}}$ cubic — DIFFERENT color block, “Hermitian, genuinely independent”

Cal: *“They’re describing two different operators, both called ‘the 0^{-+} ,’ with different color structure. If Elie’s cubic 0^{-+} is right, ‘color block same for all’ fails and the headline insight is wrong; if Lyra’s is right, Elie’s ordering loses one of its two computed ranks.”*

Status: by K472 (filed before this absorption), Elie HAS walked back the $d^{\{abc\}}=0^{-+}$ claim, accepting Lyra’s $\text{Tr}(F \cdot \tilde{F})$. The contradiction Cal flagged IS resolved by Lyra winning. But: in K471 itself, I had cited Elie’s 4308 ordering result as supporting the factorization framing while the contradiction was live. That was premature elevation. The factorization-as-SOLID claim needed the Elie-Lyra reconciliation FIRST.

Walk-back: the Lyra K471 factorization is LEAD-substantive (now with Elie-Lyra reconciliation in K472), not SOLID. Cal’s protocol requires the blind forward Δ test before SOLID status.

Catch 2 — “Confinement is algebraic, not dynamical” overclaim

Cal hit this hard, with three specific objections:

1. Circular — putting only color-singlets into the observable algebra assumes confinement (no free color); it doesn’t derive it.
2. Discards what confinement physically is — linear potential, string tension, finite-T deconfinement transition are all dynamical AND MEASURED; a singlet-only algebra reproduces NONE of them.
3. Collides with BST’s own honest state — you can’t claim confinement “solved algebraically” while interacting YM existence is named-open (W2, all weekend).

Cal’s honest version: *“BST’s observable spectrum contains only color singlets — the spectral SYMPTOM of confinement, consistent with no-free-color. That’s true and worth saying. It is NOT a derivation of confinement and NOT Paper-grade confinement content.”*

My specific over-claim in K471: *“Confinement IS the color-singlet selection rule on $H^2(D_{IV^5})$... it’s an algebraic selection rule built into how the bilinear Schwinger algebra acts on $H^2(D_{IV^5})$.”*

Walk-back: retract. The honest version is: BST’s observable spectrum contains only color singlets, consistent with no-free-color. This is the spectral symptom of confinement, not a derivation. Linear potential / string tension / deconfinement-transition dynamics are NOT addressed by the algebraic statement — they’re the interacting-content the W2 / #418 open frontiers are about.

Catch 3 — “Mass = information $\times$ spacetime-coupling” reframe banked too hard

Cal: *“An evocative reframe, not a derived relation (‘complexity = information’ is a relabel), and it leans on the substrate-as-computational thread — keep it internal, don’t bank or externalize.”*

My K471 framing: I treated “mass = (information from operator complexity) × (spacetime-block dimensionality)” as substantive substrate-architectural content connecting to BST’s substrate-as-computational thread.

Cal’s correct catch: “complexity = information” is a relabel, not a derivation. The substrate-as-computational thread is real BST internal framing (per memory MEMORY.md), but using it to elevate the spacetime- Δ observation to “mass = information” is rhetorical, not structural.

Walk-back: keep internal, do not bank or externalize. The honest statement is “*the spacetime block carries the channel-distinguishing structure (Δ); color block is universal floor*” — which IS the K471 factorization content, without the information-theoretic dressing.

Catch 4 — Selection rules for “entire hadronic spectrum” overreach

My K471 framing: “*This gives BST a substantive selection rule for the entire hadronic spectrum, not just glueballs.*”

Cal’s catch: “*The (singlet) $\otimes$ (spacetime- Δ) selection-rule idea is the genuinely good research direction (it’s the $J^{\wedge}PC$ -pattern test, oddballs included) — but credit it for glueballs, not ‘the entire hadronic spectrum.’ Mesons and baryons carry quarks; the quarkless-glue story doesn’t reach them yet.*”

Walk-back: the selection-rule framework applies to glueballs (quarkless assemblies), not the full hadronic spectrum. Mesons ($q\bar{q}$) and baryons (qqq) carry quark content; the factorization as established applies to gluonic operators only. Extending to “all hadrons” requires additional quark-content operator analysis we haven’t done. Glueball selection rules: SOLID (textbook + Cal #344 compliant). Hadronic-spectrum selection rules: not established.

The frame Cal pointed at

“Keeper’s own table marks the blind Δ forward-test as ‘still pending Elie — the Cal #344 gate.’ So by his own accounting the ground floor isn’t poured — which means the four elevations to SOLID/Paper-grade are ahead of the evidence.”

Cal is correct. K471 elevated the factorization to SOLID, banked the confinement reading, banked the mass-as-information reframe, and overreached the selection rules to the full hadronic spectrum — all BEFORE the blind forward Δ test that K471’s own routing table flagged as the gate.

The factorization framing IS a real organizing insight worth keeping. It just needs the Lyra/Elie reconciliation (now done in K472) AND the blind forward Δ test (Elie’s open #418 frontier work) before going SOLID.

Pattern flag — Cal-on-Keeper TWICE in two days

Day	Cal log	Catch
Sunday→Monday transition	Cal #344	Operator-class taxonomy as unfalsifiable post-hoc; blind-test protocol required
Monday midday	Cal #347	K471 roll-up over-tiers four things (factorization SOLID; confinement algebraic; mass=info reframe; hadronic-spectrum reach)

Pattern: Keeper at peak-convergence absorbs the team’s substantive bench work as roll-up SOLID claims faster than the protocol allows. Same FF-28 leak pattern I diagnosed for Lyra’s Paper B v0.2 abstract — bodies tier honestly, headlines/roll-ups over-elevate. Now firing on my own audits.

Methodology event candidate: “*Keeper roll-ups need their own Cal #344 protocol — tier elevations require explicit verification of the prerequisites the body already names.*” Worth Casey’s call on whether this becomes formal stack alongside the eight existing candidates.

What survives K471 + K472 at honest tier (post Cal #347 trim)

Item	Tier at K471 (overclaim)	Tier at K473 (corrected)
Lyra color×spacetime factorization	SOLID	LEAD-substantive structural — pending blind forward Δ test
Operator complexity ordering (Elie 4308)	LEAD with Elie self-walk-back	LEAD — walked back, replaced by selection-rule derived 1^{+-} -heaviest reasoning
$f^{\{abc\}}$ anti-Hermitian; $d^{\{abc\}}$ Hermitian-independent	SOLID at operator-algebra	SOLID at operator-algebra (but $d^{\{abc\}}$ is NOT 0^{-+} — $\text{Tr}(F \cdot \tilde{F})$ is; Elie K472 walk-back)
Net-compatibility lemma	SOLID at free/spectral	SOLID at free/spectral (Grace) — unchanged
OS bounded-checks (Lyra F270)	SOLID at free/spectral	SOLID at free/spectral — unchanged
Selection rules for glueballs (Lyra F274 + Elie 4310)	claimed for hadronic spectrum	SOLID for 2-gluon glueballs; reach corrected (NOT all hadrons)
“Confinement is algebraic, not dynamical”	banked as substrate-architectural insight	RETRACTED — honest version: observable spectrum contains only color singlets (spectral symptom of confinement, not derivation)
“Mass = information × spacetime-coupling”	banked as substantive content	INTERNAL ONLY — reframe not derivation; relabel of complexity
$0^{++}/0^{-+}/2^{++}$ degenerate at canonical $\Delta=4$	implicit	SOLID — selection-rule consequence; degeneracy is the gate
Bulk anomalous dimension on $H^2(D_{IV}^5)$ is the open frontier	noted as gate	CONFIRMED gate — the substantive computation; multi-step build with Elie+Lyra paired

What ships per Grace K469 honest disposition (unchanged)

- Paper A v0.1: gap $C_2 = 6$ SOLID; 0^{++} anchor 1720 MeV SOLID; cross-channel named-open structural-tier; five dictionaries tested
- Paper B v0.5: per Cal #336 trim + cold-read
- Operator-class taxonomy: candidate research direction, NOT v0.1 content
- Nuclear bridge: Paper C territory if taxonomy survives blind-test
- so(7) unification: LEAD-STRENGTHENED memory entry pending Casey ratification
- Confinement claim: NOT in Paper A (Cal #347)
- Mass=info reframe: internal only (Cal #347)
- Selection rules apply to glueballs only (Cal #347)

Cumulative discipline ledger — 51 events

To K472's 49, add: - 50. Cal #347 catches K471 four over-tiers — second Cal-on-Keeper audit in two days - 51. Keeper walks back K471 over-tiers clean (K473)

Fifty-one cumulative honest discipline events. The audit chain catching the audit chain catching the audit chain.

Casey decisions queued (unchanged + one new)

1. Paper A v0.1 ship-state per Grace K469 — confirmed disposition holds

2. Methodology candidate: “Keeper roll-ups need Cal #344 protocol applied to own tier elevations” — new from K473
3. Methodology stack: now 9 candidates (Sunday 6 + “Check the gate” K468 + “Blind class-assignment” K470 + “Roll-up tier-elevation discipline” K473)
4. so(7) memory entry pending ratification
5. Bulk anomalous Δ scoping (Elie+Lyra) — proceed
6. Confinement claim: NOT in Paper A
7. Mass=info reframe: internal only

— Keeper, 2026-06-22 Mon 12:30 EDT